Kinematic and 6-DOF Singularity Analysis Report ($q_7 = -q_1$ Constrained)

Automated Generation System

March 19, 2026

1 Mathematical Formulation & Singularity Extraction

The closed-loop spatial mechanism is driven by q_1, with the geometric symmetry explicitly maintained via the constraint $q_7 = -q_1$. To evaluate the structural singularity of the entire mechanism loop, we extract the passive Jacobian $J_p \in R^{6\times6}$ corresponding to the remaining loop kinematics $J_2 \dots J_7$.

By identifying the local minima of the 6th singular value (σ_6), the following critical singular configurations (dead centers) were numerically extracted:

- Driven by $q_1 = 312.87°$, the minimum singular value drops to $\sigma_6 = 0.000264$.

2 Example Matrix Calculation at $q_1 = 0°$

The solved full joint configuration (in radians) is:

$$\mathbf{q} = \begin{bmatrix} 0.0000 \\ 0.0009 \\ 0.0000 \\ -0.0017 \\ -0.0000 \\ -0.0009 \\ -0.0000 \end{bmatrix} \tag{1}$$

The full 6×7 Jacobian matrix J:

$$J = \begin{bmatrix} 0.000 & 0.000 & -0.000 & 2.500 & -0.002 & -2.500 & 0.000 \\ 0.000 & -1.000 & -0.000 & 3.000 & 0.000 & 2.000 & -0.000 \\ -0.000 & 0.000 & -3.000 & 0.002 & -0.500 & 0.002 & -2.000 \\ 0.000 & 0.000 & 0.000 & 0.001 & -0.000 & 0.001 & 0.000 \\ 0.000 & 0.000 & -1.000 & -0.000 & -1.000 & -0.000 & 1.000 \\ 1.000 & 1.000 & 0.000 & -1.000 & -0.000 & 1.000 & -0.000 \end{bmatrix} \tag{2}$$

The structural 6×6 passive Jacobian J_p (excluding J_1):

$$J_p = \begin{bmatrix} 0.000 & -0.000 & 2.500 & -0.002 & -2.500 & 0.000 \\ -1.000 & -0.000 & 3.000 & 0.000 & 2.000 & -0.000 \\ 0.000 & -3.000 & 0.002 & -0.500 & 0.002 & -2.000 \\ 0.000 & 0.000 & 0.001 & -0.000 & 0.001 & 0.000 \\ 0.000 & -1.000 & -0.000 & -1.000 & -0.000 & 1.000 \\ 1.000 & 0.000 & -1.000 & -0.000 & 1.000 & -0.000 \end{bmatrix} \tag{3}$$

Applying SVD on $J_p = U\Sigma V^T$ yields 6 singular values. The left singular vectors U:

$$U = \begin{bmatrix} 0.667 & -0.003 & -0.641 & 0.000 & -0.380 & 0.000 \\ 0.671 & 0.001 & 0.738 & -0.000 & -0.068 & 0.000 \\ 0.002 & 0.990 & -0.003 & 0.142 & 0.000 & -0.000 \\ 0.000 & 0.000 & 0.000 & -0.000 & -0.000 & -1.000 \\ 0.000 & 0.142 & -0.001 & -0.990 & 0.000 & 0.000 \\ -0.324 & 0.001 & 0.210 & -0.000 & -0.923 & 0.000 \end{bmatrix} \tag{4}$$

The singular values matrix Σ (6×6):

$$\Sigma = \begin{bmatrix} 4.177 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 3.669 & 0.000 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 3.355 & 0.000 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 1.669 & 0.000 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.892 & 0.000 \\ 0.000 & 0.000 & 0.000 & 0.000 & 0.000 & 0.000 \end{bmatrix} \tag{5}$$

And V^T (6×6):

$$V^T = \begin{bmatrix} -0.238 & -0.002 & 0.959 & -0.001 & -0.155 & -0.001 \\ 0.000 & -0.848 & -0.001 & -0.174 & 0.004 & -0.501 \\ -0.157 & 0.003 & 0.119 & 0.001 & 0.980 & 0.002 \\ -0.000 & 0.338 & 0.000 & 0.551 & -0.000 & -0.763 \\ -0.958 & -0.001 & -0.258 & 0.001 & -0.123 & 0.000 \\ -0.001 & 0.408 & -0.000 & -0.816 & 0.000 & -0.408 \end{bmatrix} \tag{6}$$

3 Analysis Results & Curve Fitting

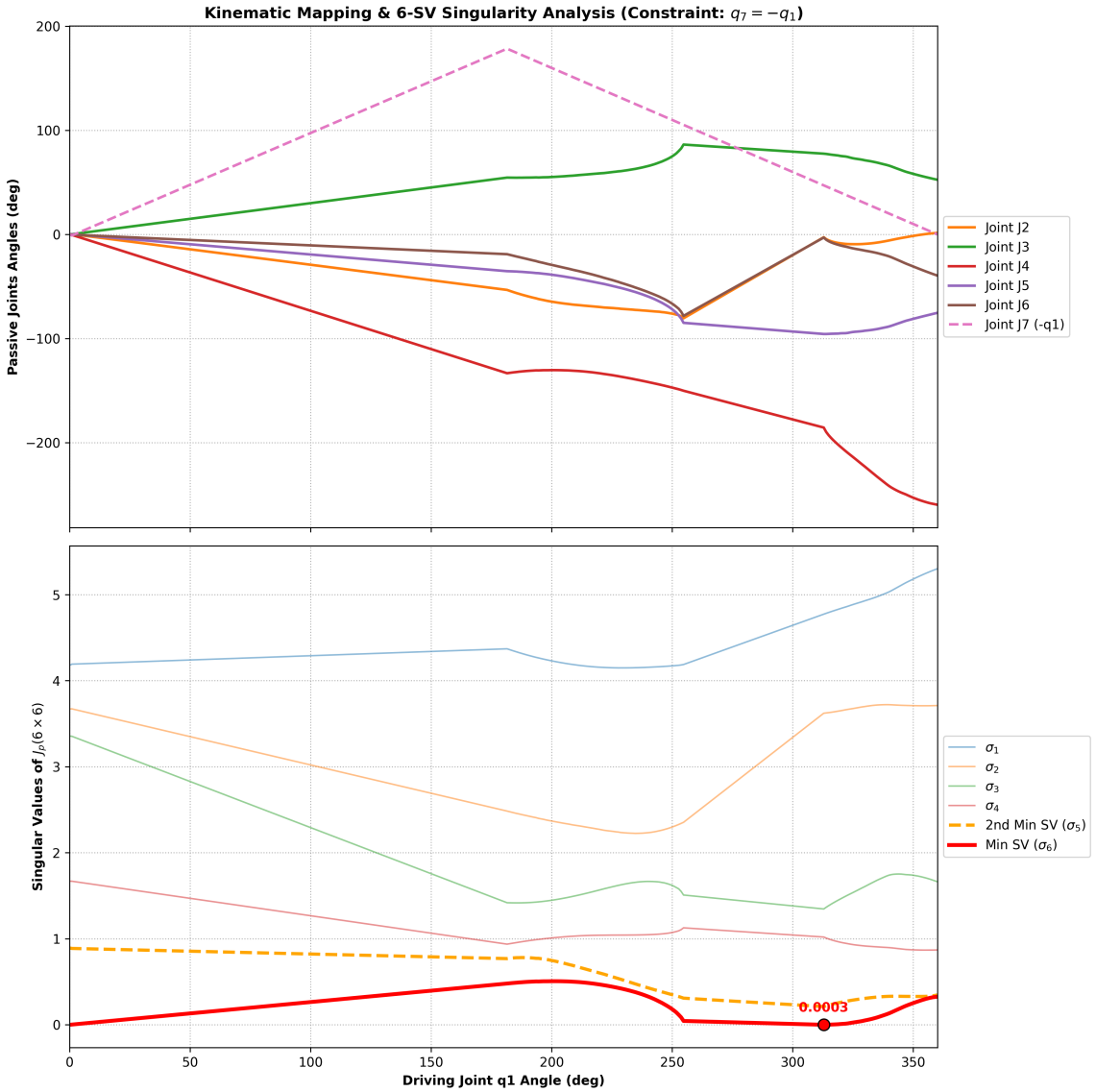


Figure 1: Kinematic mapping and full 6×6 singular value evolution. Critical dead centers are highlighted in red.

The 3rd-order Fourier series expressions for the joints are:

- **Joint J2:**
  $Q_{J2}(x) = +55.136 + 34.833\cos(x) + 166.434\sin(x) - 78.551\cos(2x) - 0.346\sin(2x) - 7.490\cos(3x) - 30.914\sin(3x)$
- **Joint J3:**
  $Q_{J3}(x) = -66.693 - 13.783\cos(x) - 225.434\sin(x) + 108.790\cos(2x) - 23.337\sin(2x) + 7.051\cos(3x) + 36.319\sin(3x)$
- **Joint J4:**
  $Q_{J4}(x) = +270.114 + 30.879\cos(x) + 719.594\sin(x) - 417.467\cos(2x) + 90.080\sin(2x) - 61.597\cos(3x) - 117.017\sin(3x)$

- **Joint J5:**
  $Q_{J5}(x) = +99.266 - 0.373\cos(x) + 283.813\sin(x) - 138.831\cos(2x) + 30.306\sin(2x) - 11.160\cos(3x) - 44.027\sin(3x)$

- **Joint J6:**
  $Q_{J6}(x) = +130.631 + 3.319\cos(x) + 283.169\sin(x) - 149.405\cos(2x) - 7.648\sin(2x) - 9.840\cos(3x) - 61.791\sin(3x)$

- **Joint J7 (-q1):**
  $Q_{J7(-q1)}(x) = +89.832 - 94.932\cos(x) + 0.138\sin(x) + 0.049\cos(2x) - 25.770\sin(2x) + 5.241\cos(3x) - 0.242\sin(3x)$